Introduction to Economics

Class 8 — Surplus, efficiency, taxes and subsidies

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Universitat de Barcelona · Group A6

19 October 2026

Quiz 5

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top.

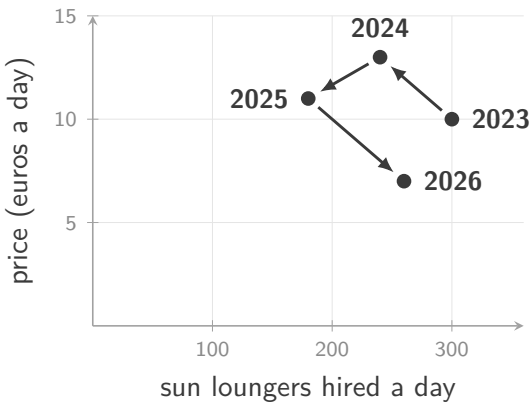
Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

10:00	Quiz 5
10:10	Warm-up: which curve moved?
10:14	Bread, milk, eggs: VAT down to zero
10:20	What buyers gain: consumer surplus
10:26	What sellers gain: producer surplus
10:29	In pairs: menús del día
10:36	Total surplus: efficient, but fair?
10:44	A tax: a wedge between two prices
10:50	Break
11:00	Who pays the tax office; where the money goes
11:08	In pairs: a tax on menús
11:15	Who bears a tax: elasticity decides; in pairs
11:29	A subsidy; VAT; the bread answered
11:40	In one line; before Thursday 22 October

Warm-up: two dots — which curve moved?



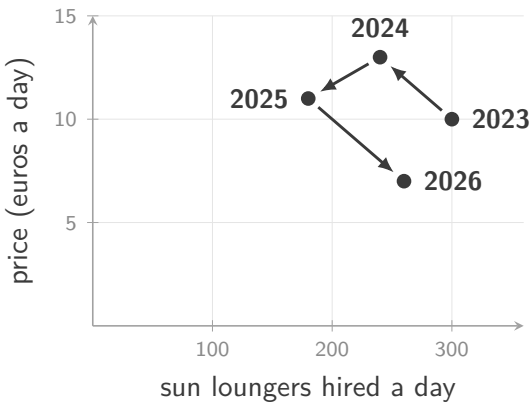
Sun loungers on one beach, each August

Each year only one curve moved. Which one, which way?

1. 2023 → 2024
2. 2024 → 2025
3. 2025 → 2026

In pairs, two minutes; then hands per arrow.

Warm-up: two dots — which curve moved?

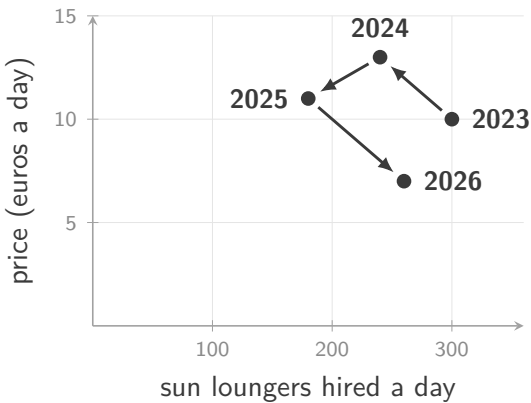


Sun loungers on one beach, each August

Each year only one curve moved. Which one, which way?

1. 2023 → 2024
   supply left: dearer, fewer
2. 2024 → 2025
3. 2025 → 2026

Warm-up: two dots — which curve moved?

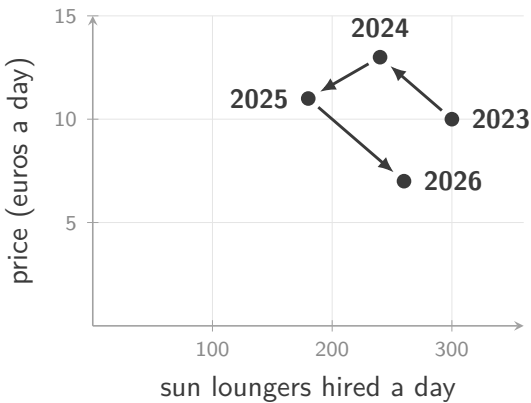


Sun loungers on one beach, each August

Each year only one curve moved. Which one, which way?

1. 2023 → 2024
   supply left: dearer, fewer
2. 2024 → 2025
   demand left: cheaper, fewer
3. 2025 → 2026

Warm-up: two dots — which curve moved?



Sun loungers on one beach, each August

Each year only one curve moved. Which one, which way?

1. 2023 → 2024
   supply left: dearer, fewer
2. 2024 → 2025
   demand left: cheaper, fewer
3. 2025 → 2026
   supply right: cheaper, more

1 January 2023: VAT on bread, milk and eggs fell from 4 % to 0 %



bread, milk, cheese, eggs, fruit, vegetables

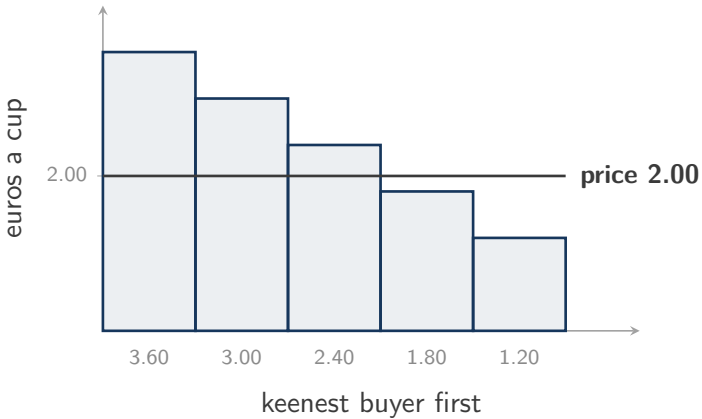


olive and seed oils, pasta

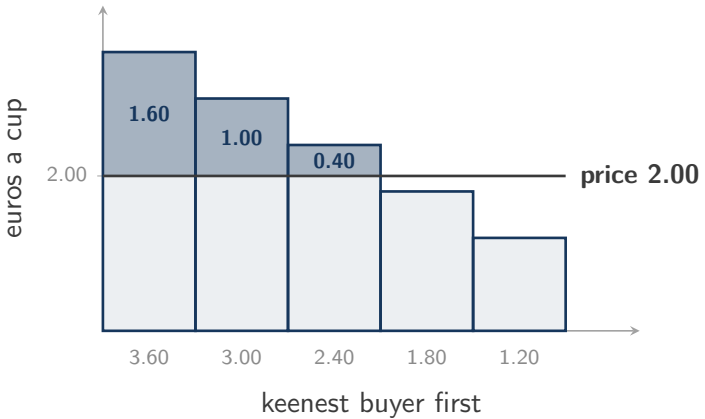
A 1.04 € loaf: 4 cents of VAT.
How many reached the shopper?

Real Decreto-ley 20/2022 (BOE, 28 December 2022), in force 1 January 2023.

Willingness to pay: each step of demand is one buyer

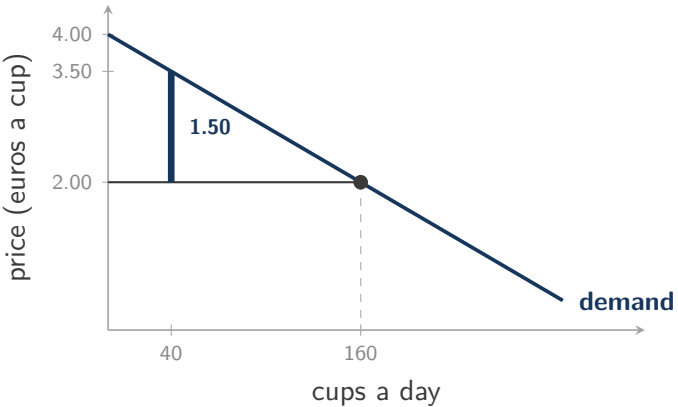


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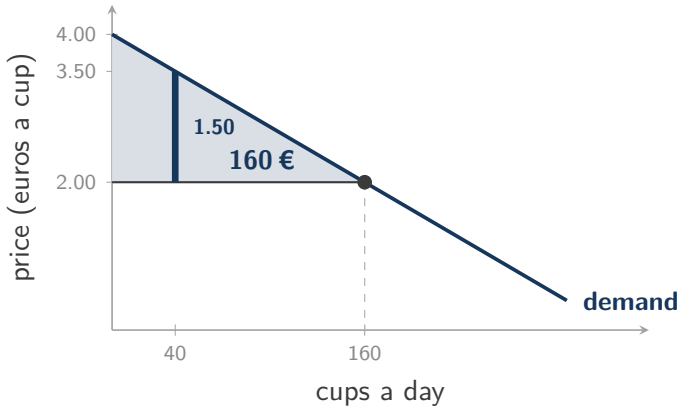


Consumer surplus = willingness to pay − price: $1.60 + 1.00 + 0.40 = 3.00\,€$.

Consumer surplus: under demand, above the price — 160 € a day

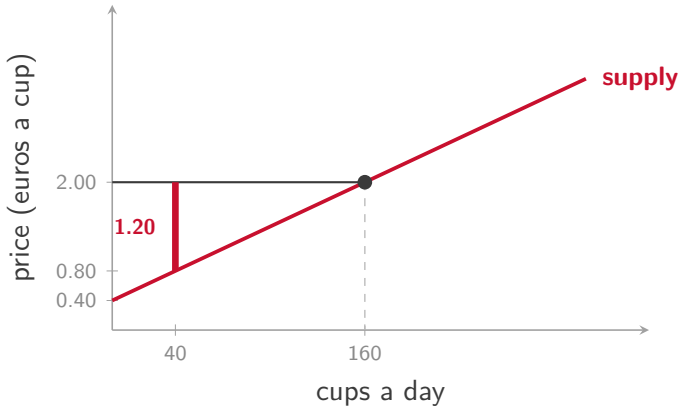


Consumer surplus: under demand, above the price — 160 € a day

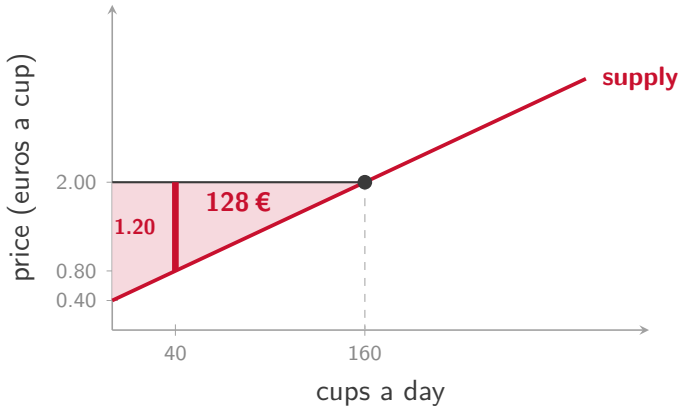


½ × 160 cups × 2.00 € = 160 euros a day: every buyer's bargain, added up.

Producer surplus: above supply, below the price — 128 € a day

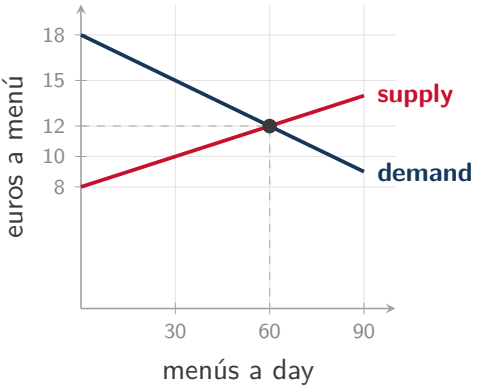


Producer surplus: above supply, below the price — 128 € a day



½ × 160 cups × 1.60 € = 128 euros a day: what the bars get above their costs.

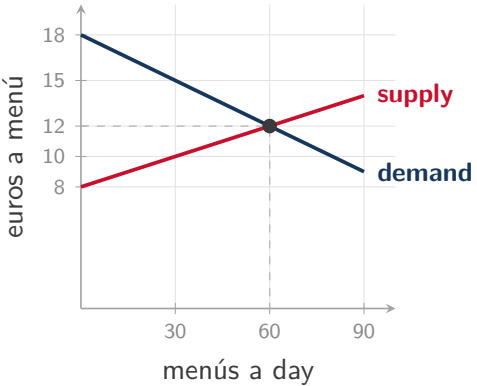
In pairs, four minutes: read the surplus off the graph



Menús del día on one street

1. Consumer surplus at 12 €?
2. Producer surplus at 12 €?
3. The 30th menú: worth? costs?

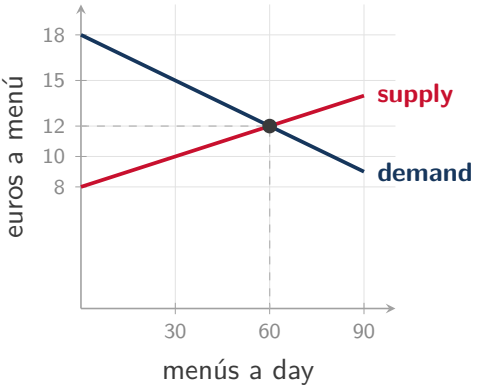
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Menús del día on one street

1. Consumer surplus at 12 €?
   buyers get 180 € a day
2. Producer surplus at 12 €?
3. The 30th menú: worth? costs?

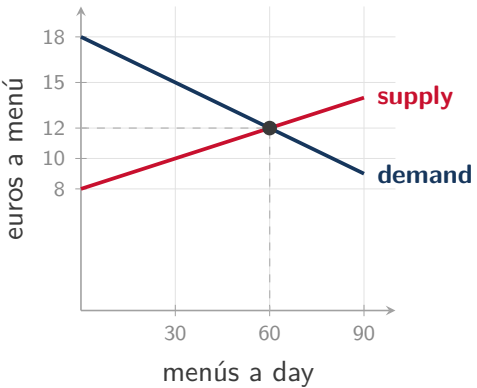
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Menús del día on one street

1. Consumer surplus at 12 €?
   buyers get 180 € a day
2. Producer surplus at 12 €?
   the bars get 120 € a day
3. The 30th menú: worth? costs?

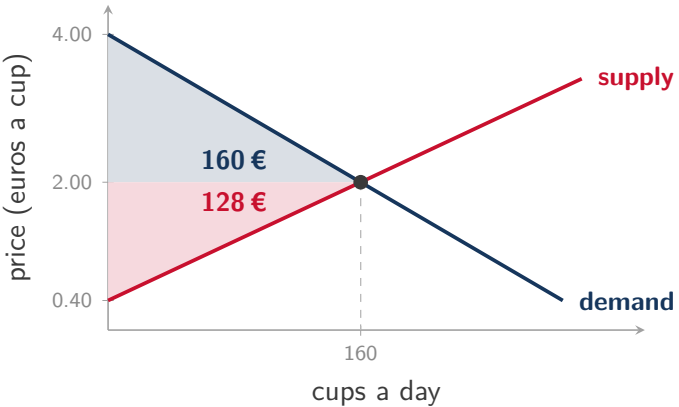
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Menús del día on one street

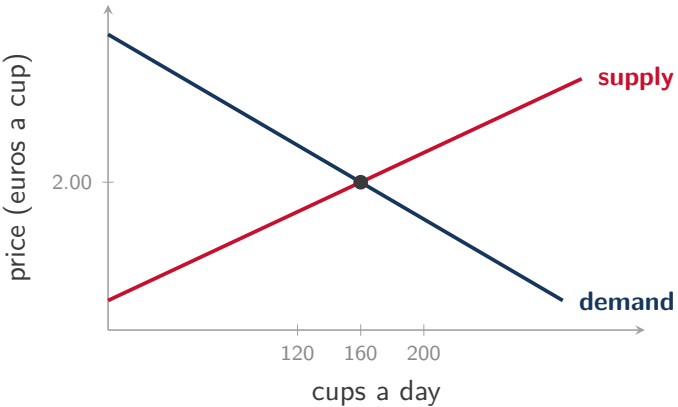
1. Consumer surplus at 12 €?
   buyers get 180 € a day
2. Producer surplus at 12 €?
   the bars get 120 € a day
3. The 30th menú: worth? costs?
   worth 15, costs 10: 5 € of surplus

Total surplus: buyers' 160 plus the bars' 128 — 288 euros a day

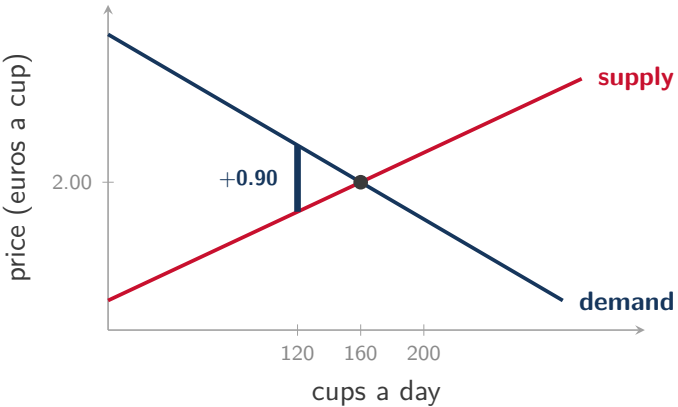


Total surplus = value to buyers − cost to sellers = 480 − 192 = 288 euros a day.

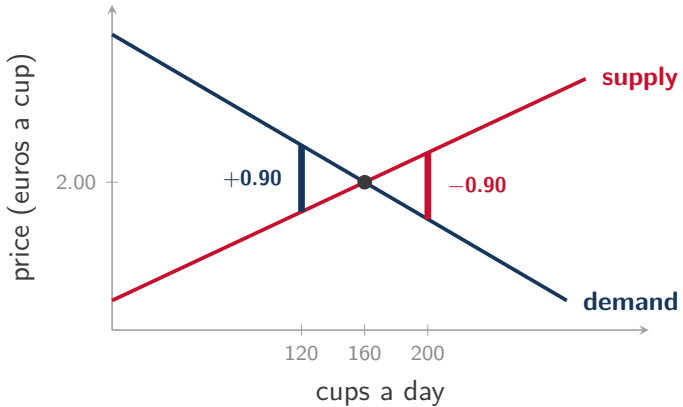
Stop at 160: before it, cups worth more than they cost; after it, less



Stop at 160: before it, cups worth more than they cost; after it, less



Stop at 160: before it, cups worth more than they cost; after it, less



Below 160: surplus left on the table. Above: cups that cost more than they are worth.

Just right: the market's 160 cups give the largest total surplus

- The cups go to those who value them most
- They are made by the bars that make them cheapest
- Every cup worth more than its cost, and no other

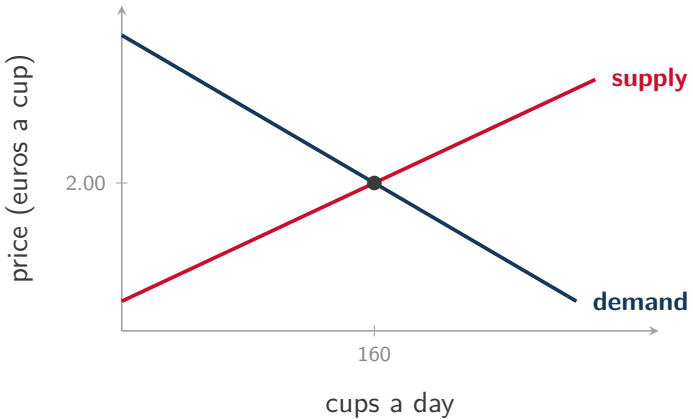
That is what economists call **efficient**, as long as no trade affects others (Class 10).

Efficient is not the same as fair

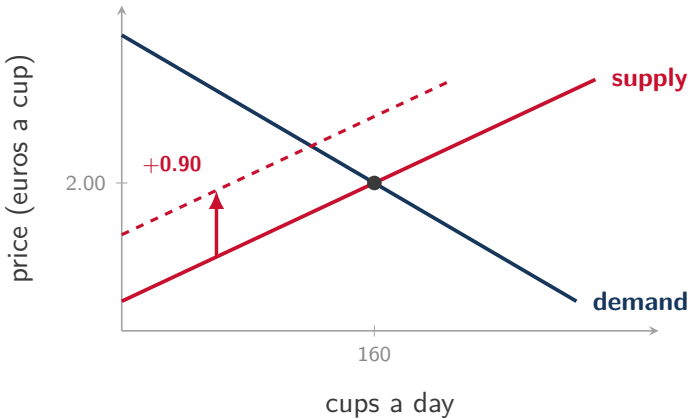
- Efficiency: is the cake as big as it can be?
- Equity: is it shared fairly?
- Willingness to pay depends on income

Efficiency is a positive question; fairness a normative one (Class 4).

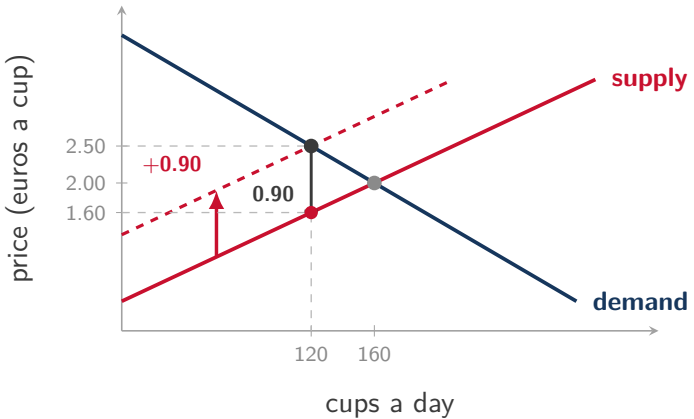
A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60



A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60

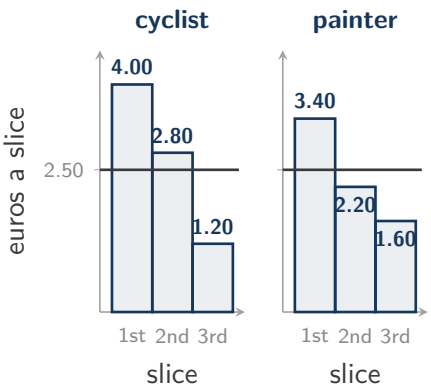


A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60



Buyers pay 0.50 more, the bars keep 0.40 less: a tax is a wedge between two prices.

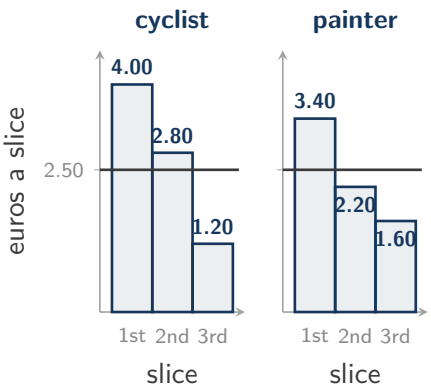
Market demand from two staircases (optional)



Pizza slices: two buyers

1. At 2.50: slices bought in all?
2. Their consumer surplus at 2.50?
3. Slices bought in all at 2.00?

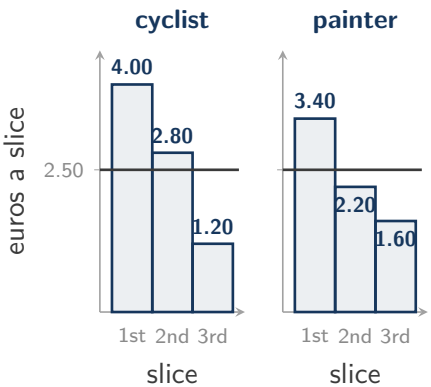
Market demand from two staircases (optional)



Pizza slices: two buyers

1. At 2.50: slices bought in all?
   3: cyclist 2 + painter 1
2. Their consumer surplus at 2.50?
3. Slices bought in all at 2.00?

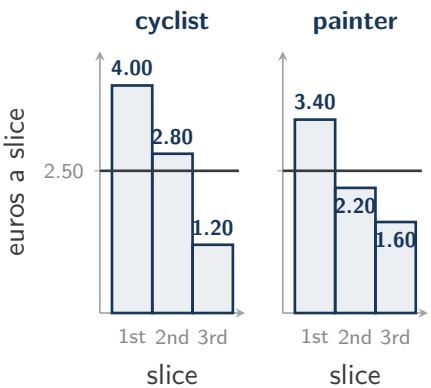
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Pizza slices: two buyers

1. At 2.50: slices bought in all?
   3: cyclist 2 + painter 1
2. Their consumer surplus at 2.50?
   1.50 + 0.30 + 0.90 = 2.70 €
3. Slices bought in all at 2.00?

Market demand from two staircases (optional)



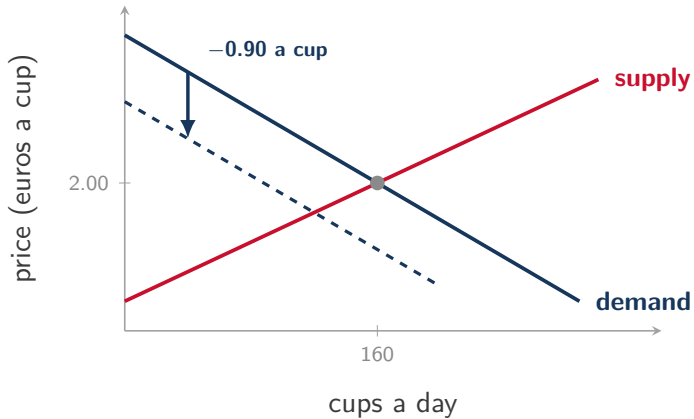
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2. Their consumer surplus at 2.50?
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3. Slices bought in all at 2.00?
   4 slices: 2 + 2

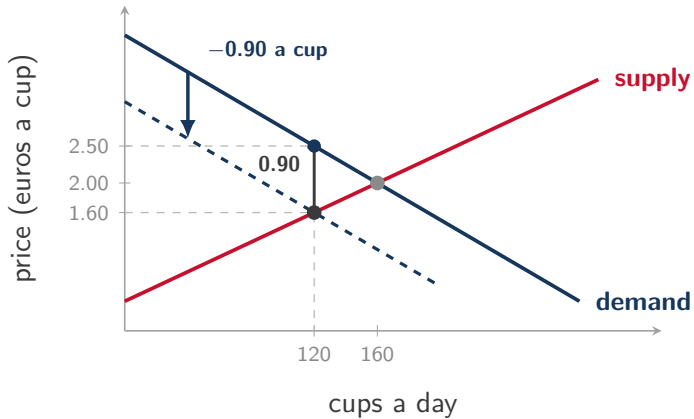
Break

Back at 11:00

Tax the buyers instead: the same 120 cups, the same 2.50 and 1.60

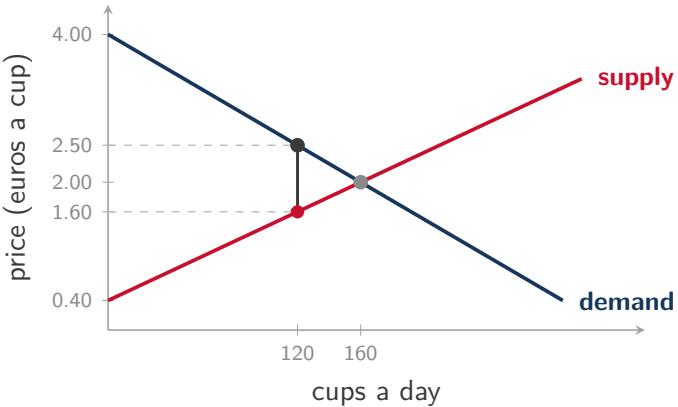


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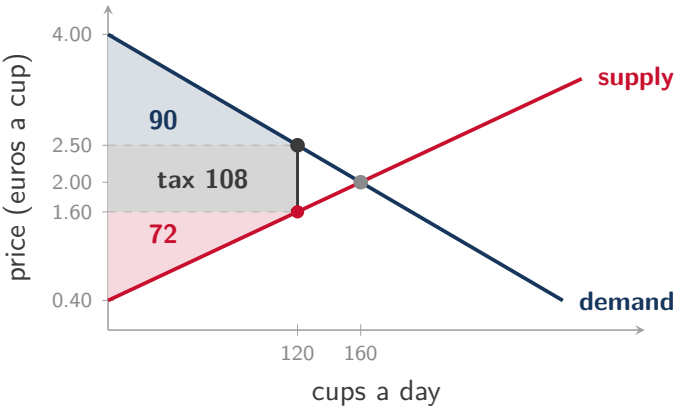


Who hands the 0.90 to the tax office does not matter: the wedge does.

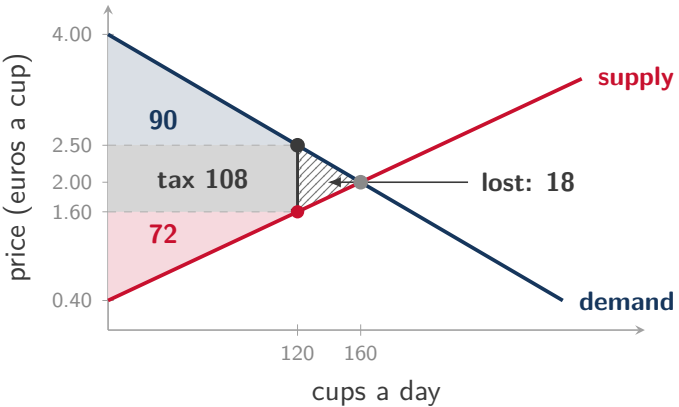
With the tax: buyers 90, bars 72, the state 108 — and 18 lost



With the tax: buyers 90, bars 72, the state 108 — and 18 lost



With the tax: buyers 90, bars 72, the state 108 — and 18 lost



Buyers lose 70, the bars 56; the state gets 108. The lost 18: the **deadweight loss**, Thursday.

In pairs, four minutes: a tax of 2.50 on every menú

Same street, same menús

menús a day	the last one is worth	the last one costs
30	15.00	10.00
45	13.50	11.00
60	12.00	12.00

1. No tax: price and menús?
2. Tax 2.50: menús, buyers pay, bars keep?
3. Tax collected a day?
4. Who bears more of the 2.50?

In pairs, four minutes: a tax of 2.50 on every menú

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1. No tax: price and menús?
   12.00 and 60 menús
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   45 menús: buyers 13.50, bars 11.00
3. Tax collected a day?
   $2.50 \times 45 = 112.50$ €
4. Who bears more of the 2.50?

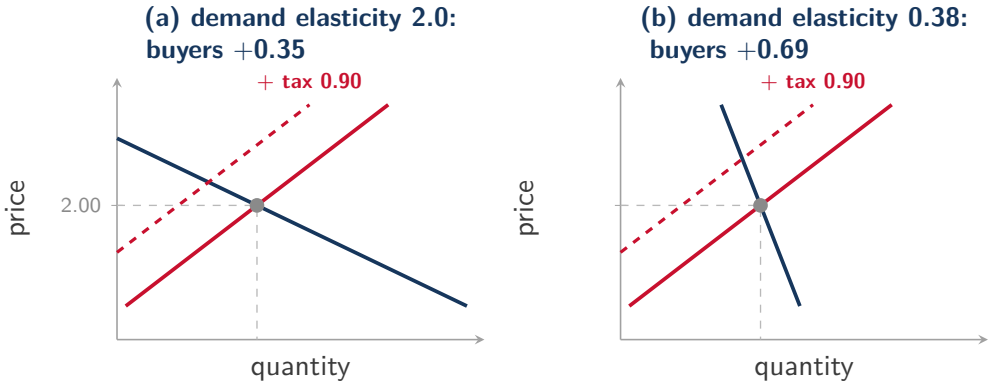
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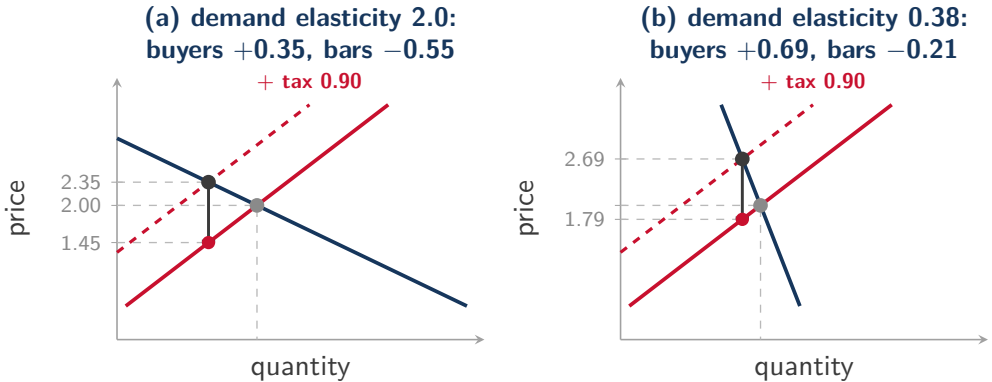
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   $2.50 \times 45 = 112.50$ €
4. Who bears more of the 2.50?
   buyers: 1.50 of the 2.50

Class 7's markets, taxed 0.90: who cannot walk away pays more



Class 7's markets, taxed 0.90: who cannot walk away pays more



Buyers who can switch leave most of the tax to the bars; buyers who cannot pay most of it.

Tax incidence: the less elastic side bears more of the tax

	most of the tax falls on
demand less elastic	buyers
supply less elastic	sellers
equally elastic	half each

The law says who pays the tax office; elasticities say who bears the tax.

In pairs, three minutes: who bears most of the tax?

Buyers or sellers — and why?

1. Cigarettes: demand elasticity 0.4, supply 2.0
2. Beach umbrellas on a July Sunday: demand 3.0, supply 0.5
3. Tax 1.00; buyers' price 5.00 $\rightarrow$ 5.80. Sellers keep? Less elastic side?
4. The same tax, paid by the buyers: the seller's price, tax not included?

In pairs, three minutes: who bears most of the tax?

Buyers or sellers — and why?

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   sellers: the umbrellas are there anyway
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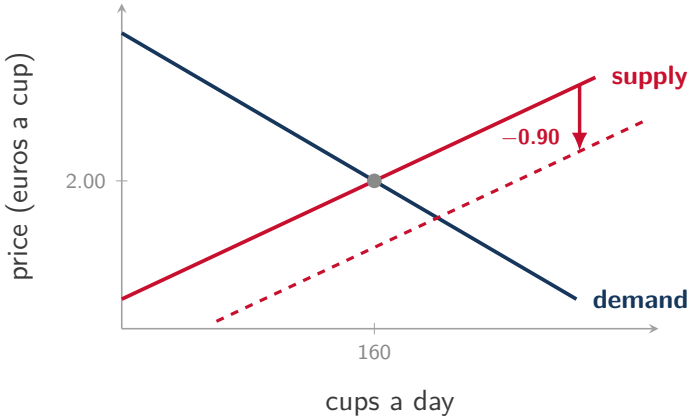
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   sellers: the umbrellas are there anyway
3. Tax 1.00; buyers' price 5.00 $\rightarrow$ 5.80. Sellers keep? Less elastic side?
   sellers keep 4.80; buyers bear 0.80: demand less elastic
4. The same tax, paid by the buyers: the seller's price, tax not included?

In pairs, three minutes: who bears most of the tax?

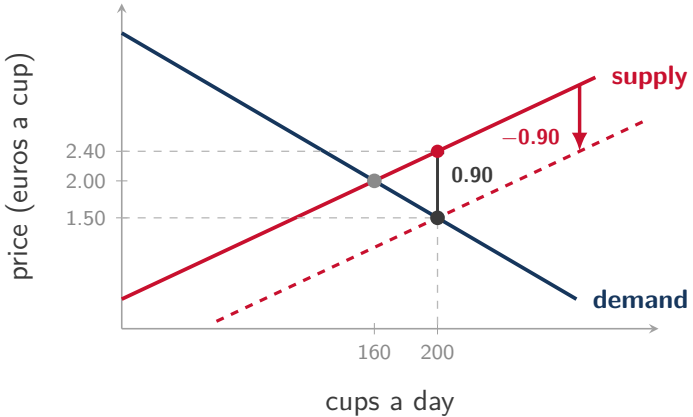
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2. Beach umbrellas on a July Sunday: demand 3.0, supply 0.5
   sellers: the umbrellas are there anyway
3. Tax 1.00; buyers' price 5.00 $\rightarrow$ 5.80. Sellers keep? Less elastic side?
   sellers keep 4.80; buyers bear 0.80: demand less elastic
4. The same tax, paid by the buyers: the seller's price, tax not included?
   4.80, tax not included; buyers pay 5.80 in all

A subsidy of 0.90 a cup: buyers pay 1.50, the bars get 2.40



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The tax reversed: buyers save 0.50, the bars gain 0.40.
The state pays $0.90 \times 200 = 180\,€$ a day.

Per unit or per cent: VAT is a percentage of the price

- A fixed sum per unit: a **specific tax** (0.90 a cup)
- A share of the price: **ad valorem** (VAT)
- Spain's VAT: 21 %, 10 %, 4 %
- A loaf: $1.00 + 4\,\%$ VAT $= 1.04\,€$

Either way a wedge; the elasticities decide who bears it.

The VAT cut on basic foods: about 90 % reached shop prices

The 4 cents of VAT on a 1.04 € loaf, split at 90 % (the figure for all the cut foods)



Inelastic buyers bore most of the VAT, so they got most of the cut.

January–February 2023, large supermarkets: over 90 % (EsadeEcPol, Brief 36, 30 March 2023).

One large chain, all foods with the cut: about 90 % by week 2 (Banco de España, Working Paper 2417, 2024).

Which curve moves? (optional)

Cinema tickets in one city

1. A tax of 1.00 a ticket, paid by the cinemas: which curve shifts?
2. Tickets get dearer: does demand shift too?
3. No tax; a subsidy makes streaming cheaper. Ticket demand?
4. And the ticket price, and the tickets sold?

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   no: a move along demand
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   shifts left: a cheaper substitute
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   supply, up by 1.00
2. Tickets get dearer: does demand shift too?
   no: a move along demand
3. No tax; a subsidy makes streaming cheaper. Ticket demand?
   shifts left: a cheaper substitute
4. And the ticket price, and the tickets sold?
   both fall: a move along supply

Who gains from trade? (optional)

Hours for one unit

	1 kg of cheese	1 bottle of wine
valley farm	2	1
hill farm	3	6

1. Who should make the cheese, and why?
2. At 1 bottle a kg: each farm's gain?
3. Hill farm at 4 h a kg, 2 h a bottle: any gain?

Who gains from trade? (optional)

Hours for one unit

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   hill: a kg costs it 0.5 bottle, the valley 2
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3. Hill farm at 4 h a kg, 2 h a bottle: any gain?
   no gain: both give up 2 bottles a kg

Surplus: value minus price for buyers, price minus cost for sellers.

The market's quantity makes total surplus as large as it can be.

A tax or a subsidy is a wedge; the less elastic side feels most of it.

Deadweight loss and price controls: Thursday.

Before Thursday 22 October

- Thursday: price controls; international trade and tariffs
- Read Mankiw & Taylor: Ch. 7, pp. 159–163 and 172–178; Ch. 17, pp. 380–390
- Problem set 1: due Thursday 22 October, on paper, in class
- Practise in Sextant: the units the course page lists



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